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Identification & Mapping of Water Slumping Fronts by Ultra-Deep Electromagnetic Logging While Drilling Technology in Lower Cretaceous Reservoirs of Adnoc Onshore

Maniesh Singh and Saif Al Arfi, ADNOC osnhore; Douglas Boyd and Nader Gerges, ADNOC; Wael Fares, Nigel Clegg, Ahmet Aki, and Emad Diab, Halliburton; Vikram Pandey, Maisoon M. Al Mansoori, Ibrahim A. Seddik, and Rathnakar Reddy, ADNOC onshore

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Abstract

ADNOC's limestone reservoirs suffer from the phenomena of injection water traveling preferentially at the top of the reservoir placing injection water above oil held there by capillary forces. Horizontal wells placed below areas of water override, cause the water above to slump unpredictably, increasing water cut and eventually killing the horizontal. Ultra Deep Directional Electromagnetic (EM) Logging While Drilling (LWD) tools provide the measurements to identify and map these water zones, improving reservoir management and design optimal well placement.

1D & 2D EM inversion modeling was conducted on two of ADNOC's largest oil producing reservoirs to evaluate the ability of an Ultra Deep Directional EM LWD Resistivity tool to identify water slumping in the presence of formation bed resistivity contrasts and predict depths of reliable detection (DOD) under various well trajectory scenarios. The inversion was run using depth of inversions up to 150 ft, the maximum expected vertical distance of tool to injection water. Modeling provided an optimized tool configuration (frequency, transmitter-receiver spacing's) to meet objectives. The inversion results further provided guidance for Geosteering, Geomapping and Geostopping decisions.

The inversion results in these reservoirs indicated that the Ultra Deep resistivity tool has a DOD of 50-150 ft to pick reservoir tops and water slumping or non-uniform waterfront boundaries. The real-time inversion will optimize landing and drilling long horizontal section to increase net pay for production and even through sub-seismic faults, measuring changes in the reservoir fluid distribution, reduce drilling risk and exceed well production life. This information will aid in updating static model with water flood areas, reservoir tops, faults and structure, designing better infill well spacing and trajectories within bypass oil regions, designing proactive and not reactive smart well completions to delay or reduce water production and ultimately extended plateau and improve ultimate recovery factor. Furthermore, it will aid resistivity mapping of underlying or overlying reservoirs for future development plans.

The encouraging results of this study confirmed to move forward with a field trial in these challenging reservoirs for better reservoir and fluid characterization and its management.

Introduction

Ultra-Deep Directional EM LWD Tool is an enhancement of current LWD directional resistivity technology to overcome existing challenges of geosteering-to-optimizing well placement, identifying far field reservoir water front movement, mapping varying water contacts and sub-seismic resolution faults. Ultra-Deep Directional LWD resistivity measurements are collected in tandem with conventional propagation resistivity to image both far and near objects. A combination of electromagnetic frequencies, multiple transmitter (Tx) to receiver (Rc) spacings provide multiple depths of investigation up to 100-150ft radius around the wellbore trajectory.

Direct interpretation of raw electromagnetic resistivity measurements is difficult and error prone, requiring in-depth knowledge of measurement sensitivities and formation parameters. Consequently, measurements are interpreted through 1D, 2D and 3D inversion modeling codes. Inversion results depend on the conductivity contrast between different mediums, stratigraphic structure and the tool's geometric position in relation to the surrounding nearby formations. Inversion adjusts formation parameters such as resistivity versus distance to match tool raw measurements. While drilling inversion modeling is commonly limited to a 1D formation vertical slices longitudinal to the borehole (Omeragic, et al. 2018, Dong, et al. 2015) to speed the delivery of information to the geosteering geologist. Advanced 2D (longitudinal & transverse) or 3D inversion requiring additionally computation resources and processing time is often delivered post well though this is changing with the availability of cloud resources. 2D & 3D inversions enable more detailed insight into reservoir structure (Bo et al. 2014, Theil et al. 2018, Bower et al. 2018, Clegg et al. 2020). Inversion modeling provides non-unique answers and must be supported by prior reservoir knowledge to judge whether the answers makes sense. This technology globally is used for:

a. GeoSteering horizontally in thin oil columns above an oil water contact.
b. GeoMapping injection well water fronts to the side of a horizontal borehole
c. GeoMapping formation structure and faults
d. Estimating formation resistivity of formations not yet penetrated by the drill bit (Alfelasi et al. 2020).

This manuscript highlights the outcome of prejob inversion study performed on Lower Cretaceous thick limestone reservoirs in Field ‘A’ & ‘B’ to provide confidence Extra Deep Direction LWD Resistivity technology is capable of mapping water slumping fronts. The information aids completion design to defer water production, improve oil recovery, enhance sweep efficiency (Balan et al. 2017) and improve simulation model predictive capabilities.

Ultra-Deep Directional Electromagnetic LWD Technology

Ultra-Deep electromagnetic (EM) tools use longer Tx to Rc spacing and lower transmission frequencies to achieve deeper depths of investigation than standard LWD EM tools (Constable et al. 2012, Tilsley-Baker et al. 2013, Wu et al. 2018). To increase spacing, the transmitter and receiver coils are separated on different subs to configure the depth of investigation specific to each drilling section (Wu at al., 2018). Transmission frequencies are also adjustable with a range between 96 kHz and 1 kHz for this tool. Increased depth of investigation allows early detection of formation and fluid boundaries in ideal cases up to ~200ft from the well bore (Wu et al. 2018). Identifying boundaries at these distances allows geostopping (calling TD of a section before a boundary is crossed), geosteering decisions (altering the well path to optimize the wells position within the reservoir) to be made early and geomapping of formation and fluid resistivities in areas that are not penetrated during drilling many meters away from the wellbore.

Historically, the data from these tools is run through one dimensional inversion algorithms, this represents the EM field as per a geological model plotted along the well bore representing changes in resistivity above and below the well bore. Changes to the inclination of the well path can be made to intercept or avoid resistivity boundaries, to optimize the well path or to map the geology and fluids for improved reservoir understanding.

Offset well data and the geological structure is used to create theoretical models representing the expected distribution of resistivity values along the well path, this is referred to as prejob inversion modelling. Running a 1D inversion algorithm for prewell modelling along the planned well path, delivers expected measurements to assess the usefulness of deploying the technology in a range of theoretical scenarios.

1D and 3D Inversion Methodology

Ultra-Deep Directional LWD Resistivity Propagation tools transmit an electromagnetic (EM) field into the formation rock surrounding the tool. Rock and fluids change the EM wave amplitude and phase as the waves travel through the rock from transmitter to receiver. Based on the EM wave phase shift and amplitude reduction a resistivity is calculated. On their own, they provide a limited understanding of the surrounding formation layer resistivities. Consequently, inversion algorithms are used to create models that represent the EM field as a series of geological units with different resistivity values. One dimensional (1D) inversion algorithms are used (Constable et al. 2012, Tilsley-Baker et al. 2013 and Wu et al. 2018) to represent changes in the formations above and below the wellbore. 1D inversions assume changes in formation and fluid resistivity only occur above and below the wellbore. Resistivity to the sides, in front and behind the tool are assumed to remain constant. Where the geology becomes more complicated and changes to resistivity to the sides or in front and behind the tool are experienced, 1D inversions can show distortions to the inversion canvas (Wilson et al 2019). This becomes a problem where major depositional, structural or fluid changes occur along the well path. Inversion algorithm advances have resulted in the release of 2D inversions (Thiel et al. 2016) in longitudinal and transverse planes, delivering inversions based on resistivity changes in front, along, sideways and behind the tool. The ultimate is 3D inversion that inverts for resistivity essentially for changes in resistivity in all directions (Wilson et al. 2019, Clegg et al. 2019 and Clegg et al. 2020). Where the geology is complex or fluid distribution is not uniform, 2D (longitudinal & transverse) inversions and 3D inversions are required to provide a better understanding of the reservoir.

Pre-Job Inversion

The prejob inversion study was conducted on proposed Extra Deep Directional LWD Resistivity trial wells located in two fields; Field A & Field B. Objectives, using all geological and petrophysical data: Determine waterfront depth of (reliable) detection (DOD) using different Tx & Rc spacings, operating frequencies and real time processing parameters. Inversions were conducted using vertical well resistivity profiles from wells located near to the proposed Extra Deep Directional technology trial wells. The objective was to evaluate tool response to water fronts while drilling through the planned well trajectory in 8-1/2" deviated and the 6" horizontal section.

Case Study 1

Field ‘A’ Overview

Field ‘A’ is located in onshore in the emirate of Abu Dhabi. The field is an elongated, faulted, double plunging anticline with the longer axis trending northeast-southwest. Reservoir ‘X’ in Field A is a thick limestone of Lower Cretaceous age. It is bounded at top and base by 30-50ft thick dense limestones. The reservoir is subdivided into layers A to D by 2-3ft thick lower porosity (< 1 mD) stylolite layers spaced at regular vertical intervals. Porosity is high and relatively constant while permeability trends from low at the base to high at the top. Peripheral water injection provides reservoir pressure support. Injection water preferentially travels in the higher permeability upper reservoir layers as the easiest path to move from injector well to producer (Figure 1a). Injection water is held in the upper layer by negative capillary pressures, a result of the rocks intermediate to slightly oil wet wettability (Cunningham and Chaliha, 2002).

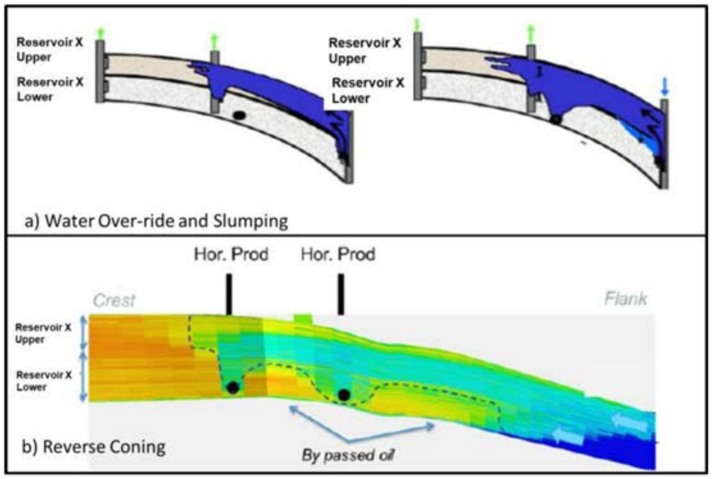

Figure 1—Water flooding & Reverse Coning mechanism in Reservoir 'X'

Producer well pressure drawdown in the lower part of Reservoir 'X' leads to reverse coning water encroachment (Figure 1b). The slumping water eventually kills the producer unless artificial lift is employed. Recovery of stranded oil between producers in the deeper reservoir layers requires close well spacings.

Well 1

Technology Trial Well1 is planned for an area where injection water encroachment and slumping is observed in the shallow layers of Reservoir 'X'. The objective of the prejob inversion study is to confirm detection of the high salinity injection water slumping fronts observed in layers A to D while drilling the 8 ½" deviated and 6" horizontal well borehole sections using selected offset well data. The prejob inversion results are documented below.

Prejob Inversion Modelling Using Offset Well 1-01V

The resistivity logs of offset well 1-01V shows a gradational resistivity profile and low resistivity contrast through Reservoir 'X's layers. Identifying specific boundaries is expected to be difficult as seen in Figure 2.

The Ultra-Deep resistivity modelling used 2 receiver subs for the both 8 ½" (landing section) and 6" (horizontal section) (Figure 3). The spacing between transmitter and receiver sub 1 is 75 feet while the spacing between transmitter and receiver sub 2 is 135 feet. The operating frequency combination used for the modeling was 4 & 8 kHz in short spacing and 2 & 4 kHz for long spacing. The depth of inversion delivered was 175 feet (up and down).

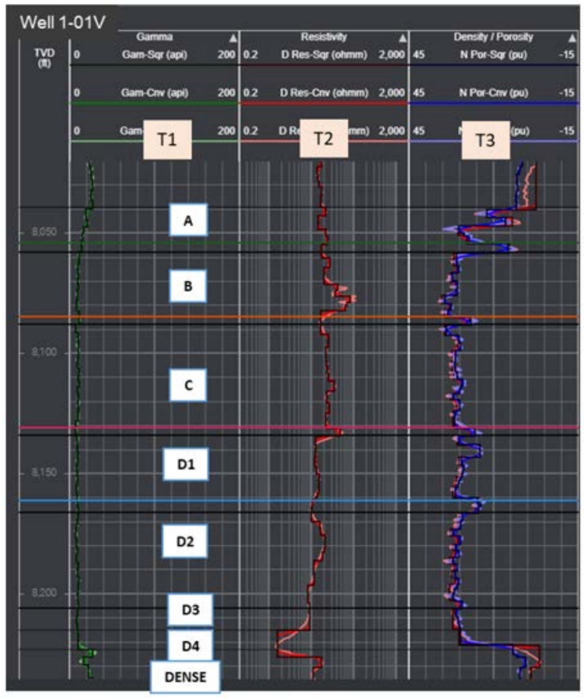

Figure 2—Basic logs of offset well 1-01V (T1: Gamma Ray, T2: Resistivity, T3: Density-Neutron).

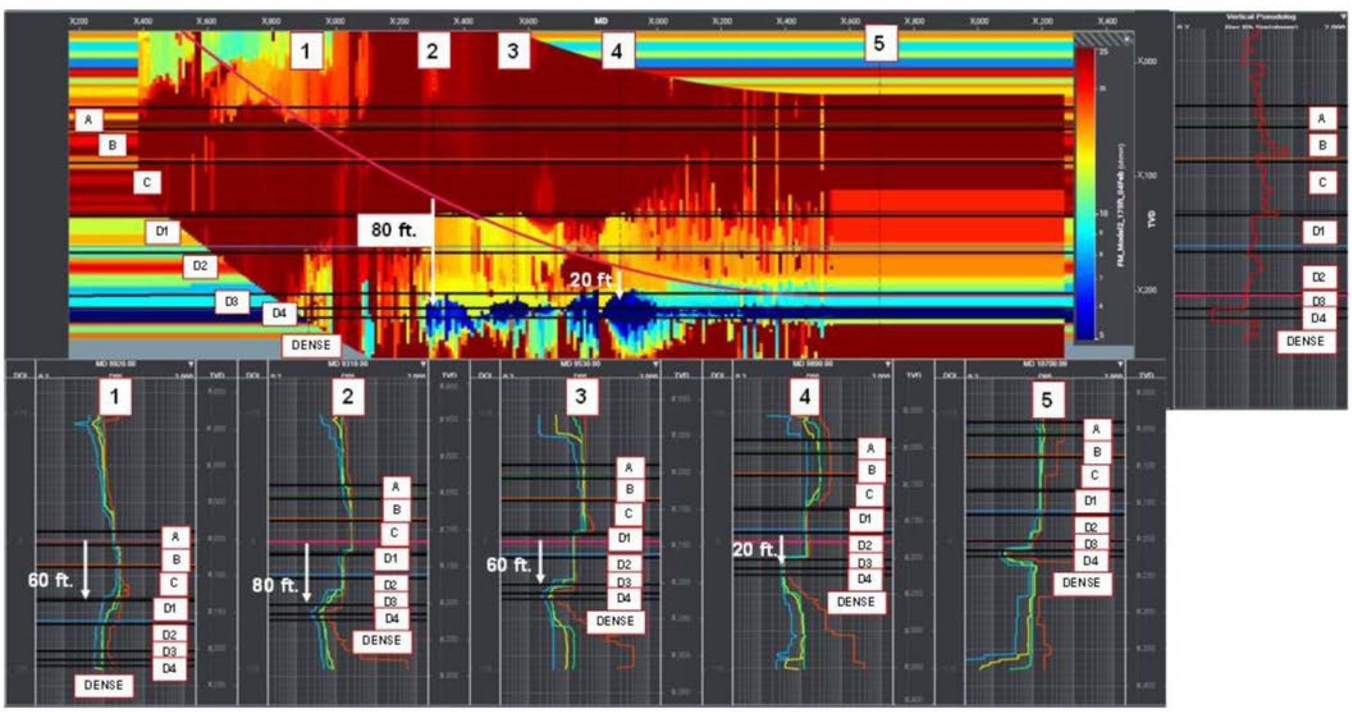

Figure 3—Stochastic inversion canvas across the 8-1/2" landing and 6" horizontal hole sections using data from offset well 1-01V. Vertical logs represent slices through the canvas.

In 8 ½" hole modelling section, the resistivity decline in reservoir layer D4 (below target reservoir) is detected 80 ft TVD below the wellbore while travelling in layer C at an inclination of 85°. In the 6" horizontal borehole modelling section, the inversion couldn't detect boundaries due to the low resistivity contrast available (Figure 3).

Prejob Inversion Modelling Using Offset Well 2-01V

The logs of offset well 2-01V display varying resistivity signatures through the reservoir layers as well as a low resistivity high salinity injection water encroachment profile in reservoir layer A. It is also observed that there is a low resistivity contrast between target reservoir D3 and overlying reservoir D2 while there is good resistivity contrast between target reservoir D3 and underlying reservoir D4 (Figure 4).

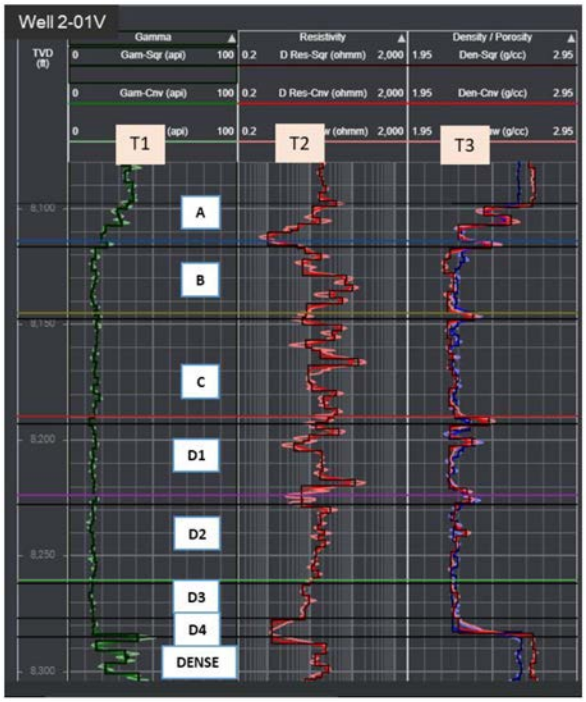

Figure 4—Basic Logs of offset well 2-01V (T1: Gamma Ray, T2: Resistivity, T3: Density-Neutron).

The Ultra-Deep resistivity modeling used 2 receiver subs for the both 8 ½" (landing section) and 6" (horizontal section) (Figure 4). The spacing between transmitter and receiver sub 1 is 50 feet while the spacing between transmitter and receiver sub 2 is 100 feet. The operating frequency combination used for the modeling was 4 & 8 kHz in short spacing and 2 & 4 kHz for long spacing. The depth of inversion was 150 feet (up and down).

In the 8 ½" hole modeling section, the significant resistivity decrease in reservoir layer A is detected 40ft TVD below from wellbore while traveling in dense limestone. While travelling in layer C at an 85° wellbore inclination layer A is 'seen' 60 ft TVD above from wellbore while D3 can be detected 110 ft TVD below. In the 6" horizontal borehole modeling section, the average canvas is detecting the significant decrease in resistivity due to water encroachment and slumping in layer B 140 ft TVD above from the wellbore travelling in layer D3 (Figure 5).

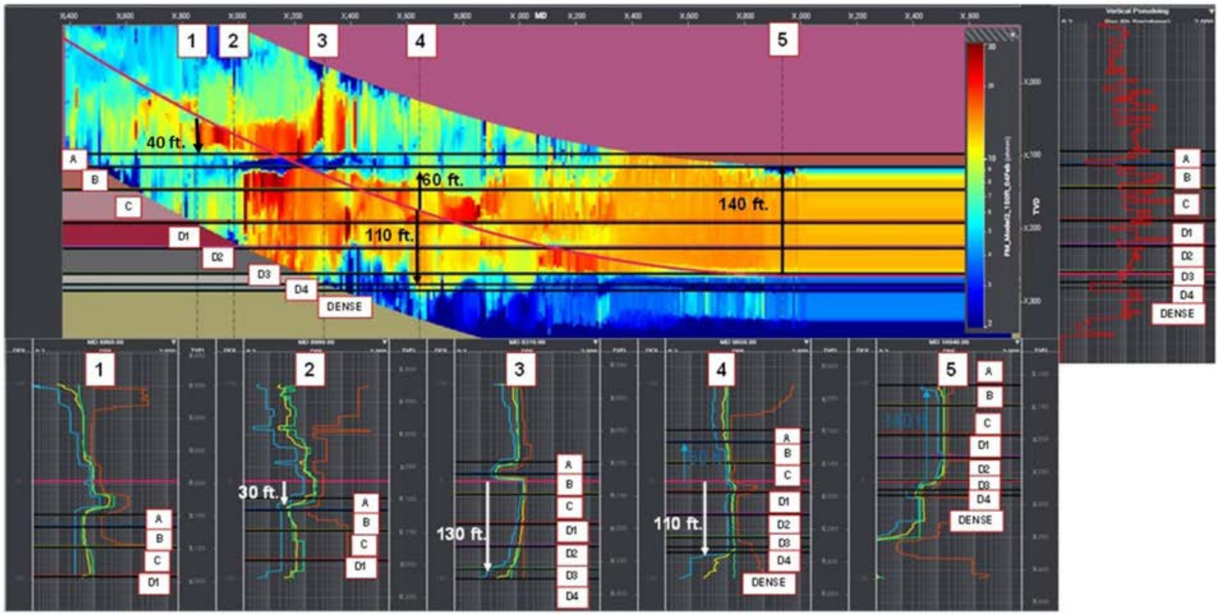

Figure 5—Stochastic inversion canvas across the 8-1/2" landing and 6" horizontal sections using offset well 2-01V. Vertical logs represent slices through the canvas.

Prejob Inversion Modelling Using Offset Well 3-01V

The resistivity log response of offset well 3-01V is similar to offset well 2-01V. It displays varying resistivity signature through the reservoir layers as well as a significant lower resistivity injection water encroachment profile in layer B. There is a low resistivity contrast between target reservoir D3 with the overlaying reservoir D2, while there is a visible resistivity contrast between target reservoir D3 and underlaying reservoir layer D4 (Figure 6).

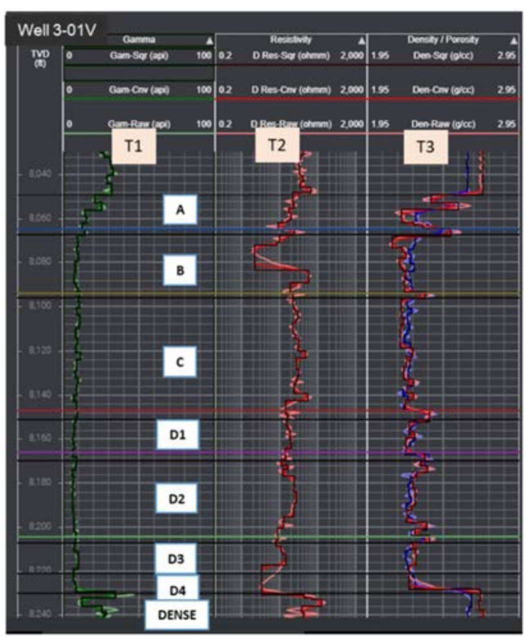

Figure 6—Basic Logs of offset well 3-01V (T1: Gamma Ray, T2: Resistivity, T3: Density-Neutron).

The Ultra-Deep resistivity modeling used 2 receiver subs for the both 8 ½" (landing section) and 6" (horizontal section). The spacing between transmitter and receiver sub 1 is 50 feet while the spacing between transmitter and receiver sub 2 is 100 feet. The operating frequency combination used for the modeling was 4 & 8 kHz in short spacing and 2 & 4 kHz for long spacing. The depth of inversion was 150 feet (up and down).

In the 8 ½" hole modeling section, the significant layer B resistivity decrease due to water advancement is detected 40 ft TVD below from the tool while travelling in dense limestone rock. While travelling in layer C at an inclination of 80°, layer D1 is identified 30 ft TVD below and layer B 60 ft above. In the 6" horizontal borehole modeling section, the tool identifies the significant resistivity decrease in layer B due to water encroachment 110 ft TVD above the wellbore in layer D3 (Figure 7).

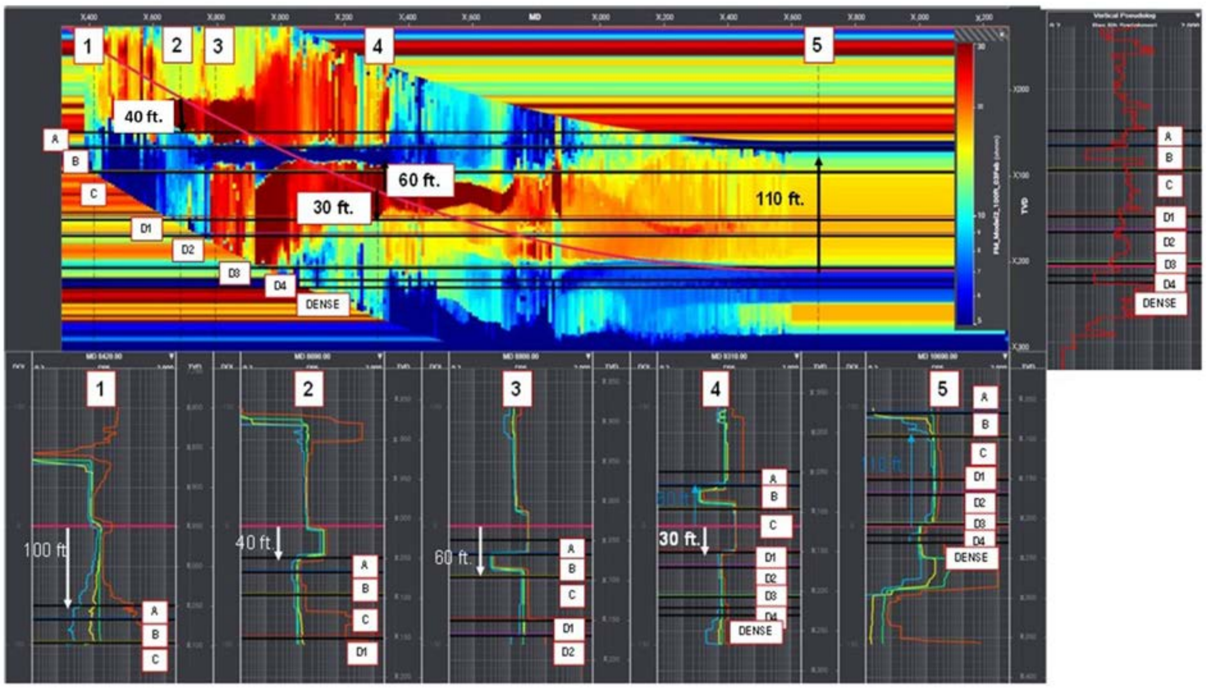

Figure 7—Inversion canvas across the landing and horizontal sections. Offset well 3-01V. Vertical logs represent slices through the canvas.

It is very evident from the 6" hole prejob inversion results of offset wells 2-01V & 3-01V that the depth of reliable detection (DOD) for water encroachment front is 110-140 ft TVD above the planned wellbore depth and trajectory if similar resistivity profiles and contrasts are encountered in the real case. This provides confidence Ultra Deep LWD EM technology can map the presence of injection water at a distance above producers while drilling allowing refinement of the completion design to defer water slumping, delaying water production.

Well 2

Trial Well2's drilling location is planned in the down flank side of Field 'A' where aggressive water slumping is observed covering the upper and middle part of reservoir 'X' as shown in the basic log response of offset wells (Figure 8). The objective of the prejob inversion study is to confirm detection of injection water encroachment fronts seen in shallow reservoir layers and identify bypass oil present in lower part of reservoir 'X', while drilling the 8 ½" and 6" section in planned trajectory. Offset vertical wells 4-01V & 5-01V display similar responses to water slumping fronts.

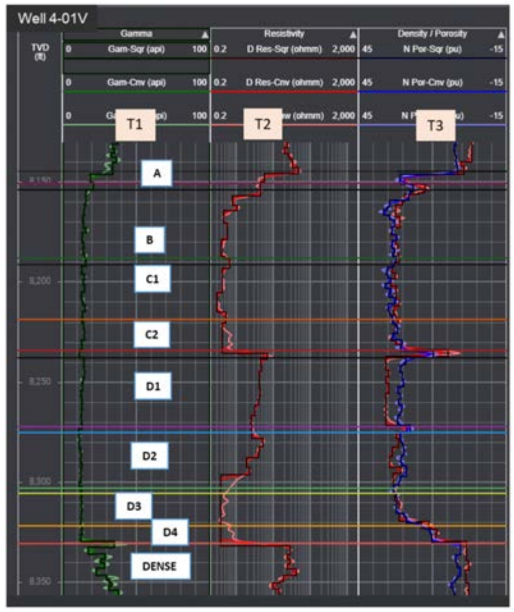

Figure 8—Basic Logs of offset well 4-01V (T1: Gamma Ray, T2: Resistivity, T3: Density-Neutron).

Prejob Inversion Modelling Using Offset Well 4-01V

Well 4-01-V log response shows declining resistivity due to injection water encroachment and slumping in reservoir layers B, C1 and C2. Elevated in resistivity in layers D1 and D2 indicates oil. The resistivity drop in layers D3 & D4 reflects the original down flank saturation transition zone (Figure 8).

Ultra-Deep resistivity modeling was conducted using 2 receiver subs for the both 8 ½" (landing section) and 6" (horizontal section) (Figure 9). The spacing between transmitter and receiver sub 1 is 50 feet while the spacing between transmitter and receiver sub 2 is 100 feet. The operating frequency combination used for the modeling was 4 & 8 kHz in short spacing and 2 & 4 kHz for long spacing. The depth of inversion was 100 feet (up and down).

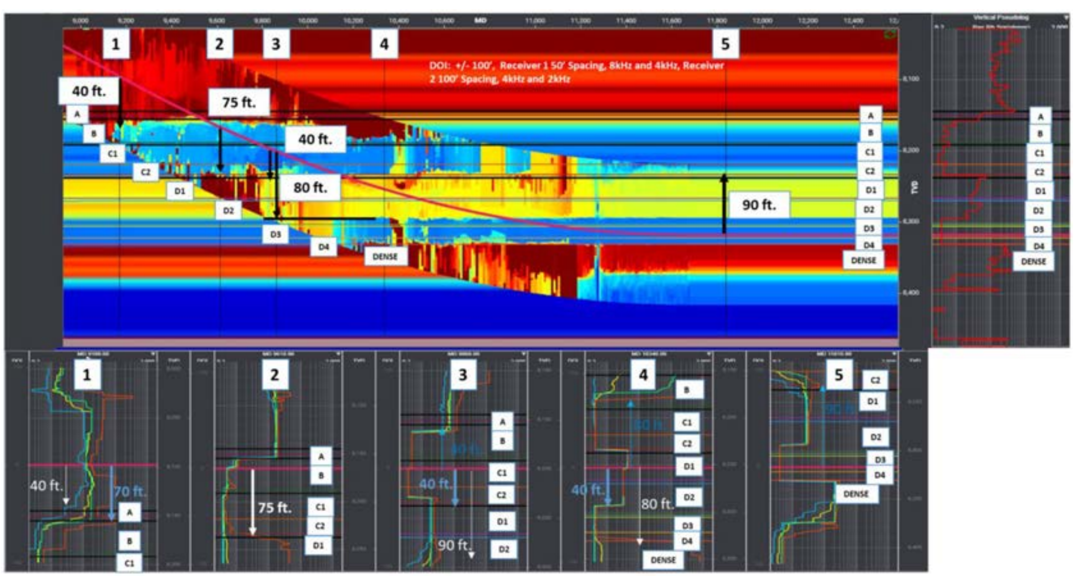

Figure 9—Inversion canvas across the 8-1/2" landing and 6" horizontal sections using offset well 4-01V. Vertical logs represent slices through the canvas.

In the 8 ½" borehole modeling section, the significant resistivity decrease in layer A is detected 40 ft TVD below the tool while traveling in dense limestone rock. While passing through conductive layer B, oil is detected in layer D1 75ft TVD below the wellbore. While travelling through conductive layer C1, top of oil in layer D1 is detected 40ft TVD below while the bottom of the D2 oil zone is detected 80ft TVD below the highly deviated wellbore. With the 6" horizontal borehole placed in conductive layer D3, the inversion canvas is detecting a significant drop in resistivity boundary due to water encroachment and slumping in layer C2 90 ft TVD above the wellbore (Figure 9).

Boundaries are expected to be detected a significant distance from the wellbore, with the approach to the target layer D3 identified 65 feet TVD below the wellbore and over 310 feet MD before it would be penetrated. The high resistivity zone below the target reservoir can be identified much earlier, 50 feet TVD below the wellbore (Figure 10).

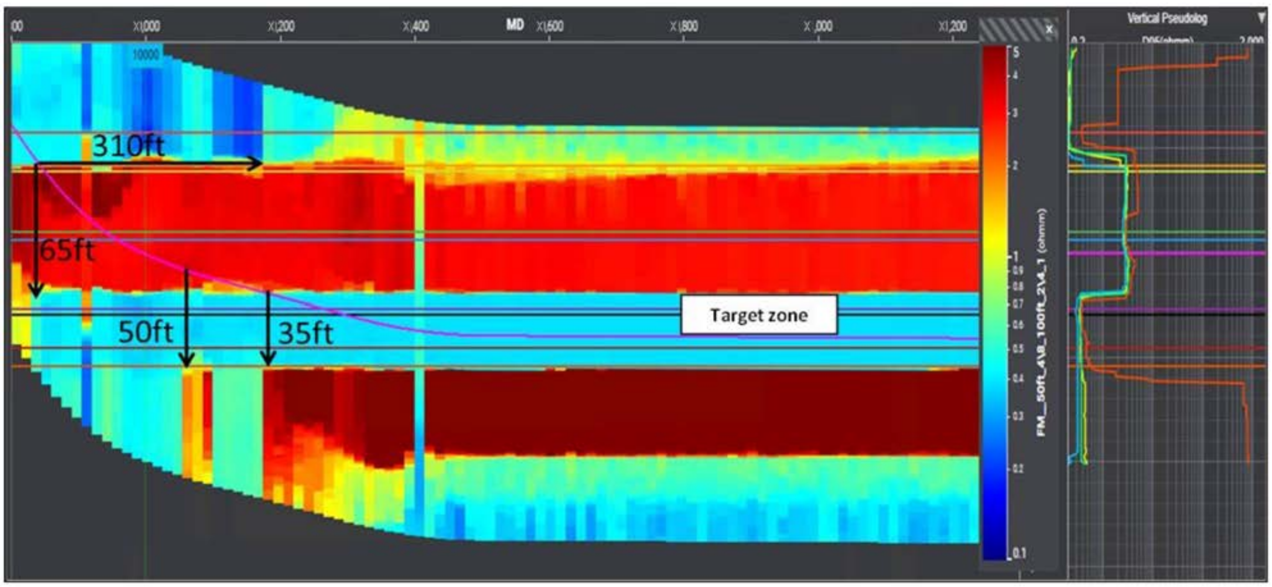

Figure 10—Inversion canvas across the landing and horizontal sections using offset well 4-01V.

Prejob Inversion Modelling Using Offset Well 5-01V

The log response of the offset well 5-01V displays a significant resistivity decline across layer B due to injection water encroachment. A low resistivity contrast exists between target layer D3 with D2 overlying layer. A large resistivity contrast exists between layer D3 and the underlying D4 layer (Figure 3).

The Ultra-Deep resistivity modeling was conducted using 2 receiver subs for the both 8 ½" (landing section) and 6" (horizontal section) (Figure 12). The Ultra-Deep resistivity inversion used 2 combinations of parameters: the first modeling used the spacing between transmitter and receiver sub 1 at 50 feet and the spacing between transmitter and receiver sub 2 at 100 feet, while the second modeling used the spacing between transmitter and receiver sub 1 at 75 feet and the spacing between transmitter and receiver sub 2 at 135 feet. The operating frequency combination used for the inversion modeling was 4 & 8 kHz in short spacing and 2 & 4 kHz for long spacing. The depth of inversion was 100 feet and 200 feet (up and down).

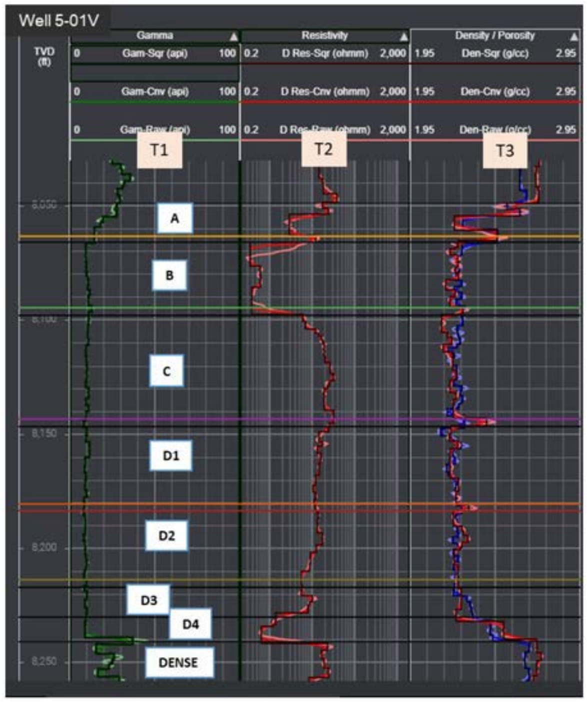

Figure 11—Basic Logs of offset well 5-01V (T1: Gamma Ray, T2: Resistivity, T3: Density-Neutron).

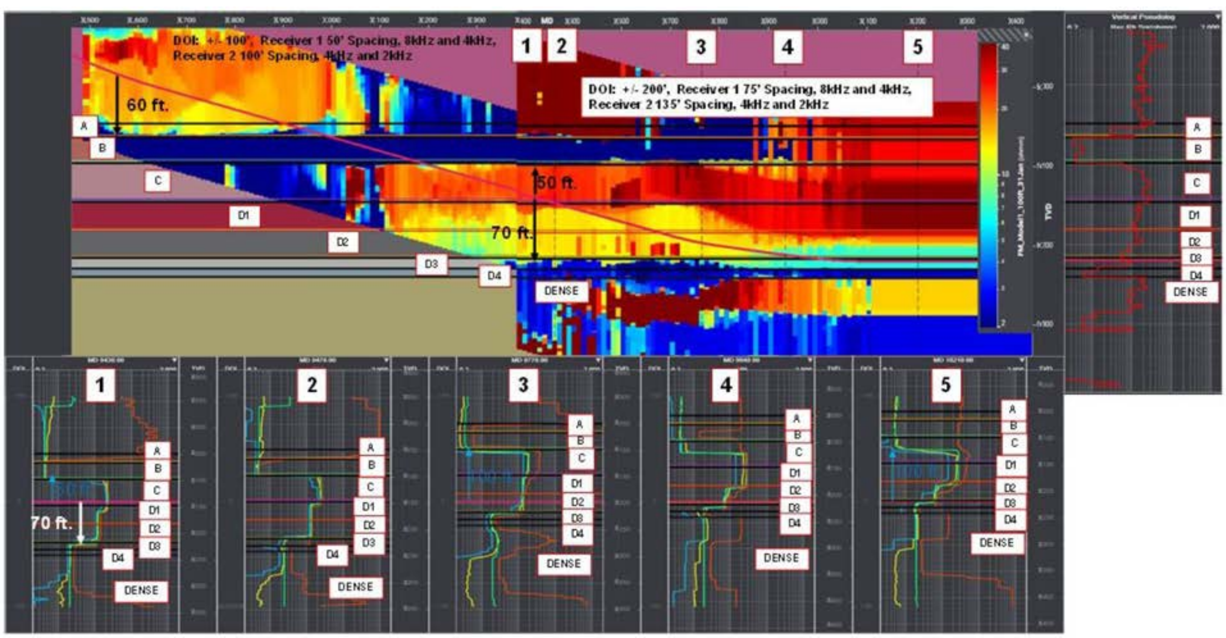

Figure 12—Inversion canvas across the 8-1/2" landing and 6" horizontal sections using offset well 5-101V. The logs represent vertical slices.

In the 8 ½" modeling section, the significant resistivity decrease in layer B is detected 60 ft TVD below the 80° inclination borehole while traveling through dense limestone. With the borehole located in layer C, the inversion identifies the significant resistivity decrease in the above layer B approximately 50 ft TVD away, meanwhile the target layer D3 is detected approximately 70 ft. TVD below the wellbore. Detecting boundaries at this distance provides information to revise the well plan to avoid landing in the low resistivity zone (Figure 12). In the 6" horizontal modeling section, the tool is detecting the significant decrease in resistivity due to water encroachment in layer B 100 ft TVD above from the wellbore located in layer D3 (Figure 12).

Case Study 2

Field B Overview

Field ‘B’ is a giant carbonate limestone oil field located onshore in Abu Dhabi emirate, U.A.E. Discovered in 1962, the field consists of three main reservoirs: Reservoir ‘R’, ‘S’ & ‘T’. The reservoirs are porous limestones separated by dense limestones and shale. Reservoir ‘R’ rests on top, separated from the underlying Reservoir ‘S’ by 45 ft of dense. Below Reservoir ‘S’ lies Reservoir ‘T’, separated by thick shale of several 100's of feet.

Structural features of the area are characterized by the occurrence of a series of NNE-SSW trending anticlines and synclines, which are probably related to deep-rooted (basement) block faults forming a series of horsts and grabens which clearly seen in seismic section in central area of field. The result of the uplift was the development of low relief anticline having an N-S structural axis. This N-S structural axis remained to the present day, but the field structural relief increased significantly, particularly along the northern and eastern flanks of the field and resulted in complete closure of the structure.

Reservoir ‘R’ has a complex internal architecture with a platform interior area developed in the southern part, a prograding platform margin in the central region, a seismically well-defined clinoform belt to the north of the central region and a basinal area in the northern extremes of the field. Reservoir ‘R’ is Aptian (Lower Cretaceous) in age and is extensively developed across the Arabian Platform. It forms the upper part of the Thamama Group, underlain by limestones and alternating limestones / marls of the adjacent Reservoir ‘S’ & ‘T’. Reservoir ‘R’ in this case study averages 300-400ft thickness and is overlain by more than 400-500ft of shales / argillaceous limestones.

Field ‘B’ is under production for many decades and is supported by water injection in Reservoir ‘R’. Injected water advances through high permeability streaks in layers V to Z.

Well 3

Trial Well3 is planned in an area where injection water encroachment occurs in high permeability V to Z layers of Reservoir ‘R’. The objective of the prejob inversion study is to confirm detection of water fronts and elevated water saturations in layers (V to Z) of Reservoirs ‘R’ while drilling. Nearby wells 6-01V, 7-01V, 8-01V were used. The trial well is planned as an 8-1/2" borehole deviated section to evaluate layers V to W and the 6" horizontal borehole section in target layer Y. The prejob inversion results of the offset wells are illustrated below.

Prejob Inversion Modelling Using Offset Well 6-01V

Offset well 6-01V shows a significant low resistivity injection water encroachment in the layer W and X and a small drop at the top of the layer Z, but otherwise shows a gradational decrease in resistivity through the stratigraphy from layer Y to Z.

The Ultra-Deep resistivity modeling was conducted using 2 receiver subs for the both 8 ½" (landing section) and 6" (horizontal section) (Figure 14). The spacing between the transmitter and receiver sub 1 is 75 feet while the spacing between transmitter and receiver sub 2 is 135 feet. The operating frequency combination used for the modeling was 4 & 8 kHz in short spacing and 2 & 4 kHz for long spacing. The depth of inversion was 200 feet (up and down).

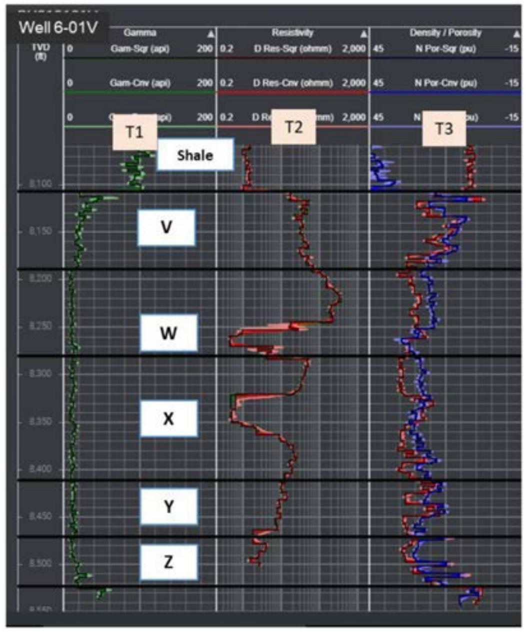

Figure 13—Basic Logs of offset well 6-01V (T1: Gamma Ray, T2: Resistivity, T3: Density-Neutron).

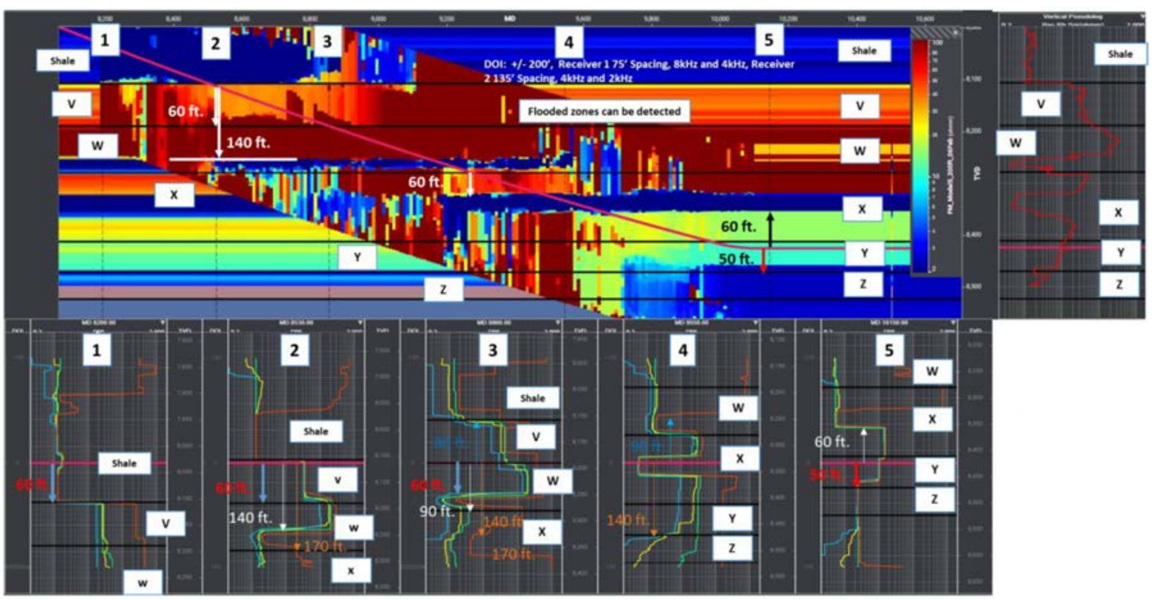

Figure 14—Inversion canvas across the 8-1/2" landing and 6" horizontal sections using offset well 6-01V. Logs represent vertical slices.

In the 8 ½" borehole modeling section while the high inclination wellbore is located in the shale formation, the top of layer W can be detected 60 ft TVD below. The water flooded zone in layer W can be detected approximately 140 ft TVD below. The water flooded zone in layer X can be detected 60 ft TVD below once the well enters resistive layer W. In the 6" borehole modeling section, the water flooded zone in layer X can be detected 60 ft up and flooded zone in layer Z can be detected 50 ft TVD below when the wellbore path is located in layer Y (Figure 14).

Prejob Inversion Modelling Using Offset Well 7-01V

The offset well 7-01V shows a gradational decrease in resistivity through the stratigraphy lower resistivity zone in layer Z. Other than layer Z, there is no specific interval with injection water encroachment creating low resistivity.

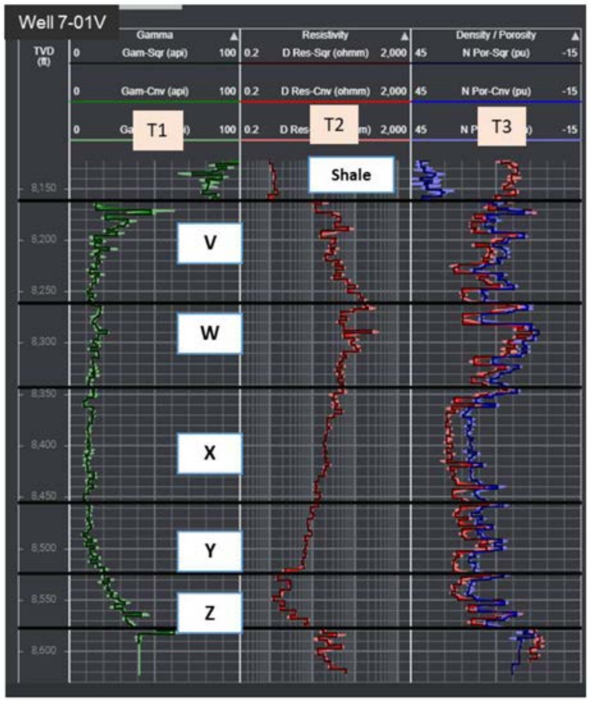

Figure 15—Basic Logs of offset well 6-01V (T1: Gamma Ray, T2: Resistivity, T3: Density-Neutron).

The Ultra-Deep resistivity modeling was conducted using 2 receiver subs for the both 8 ½" (landing section) and 6" (horizontal section) (Figure 16). The spacing between transmitter and receiver sub 1 is 50 feet while the spacing between transmitter and receiver sub 2 is 100 feet. The operating frequency combination used for the modeling was 4-8 kHz in short spacing and 2-4 kHz for long spacing. Inversion depth was 100 ft.

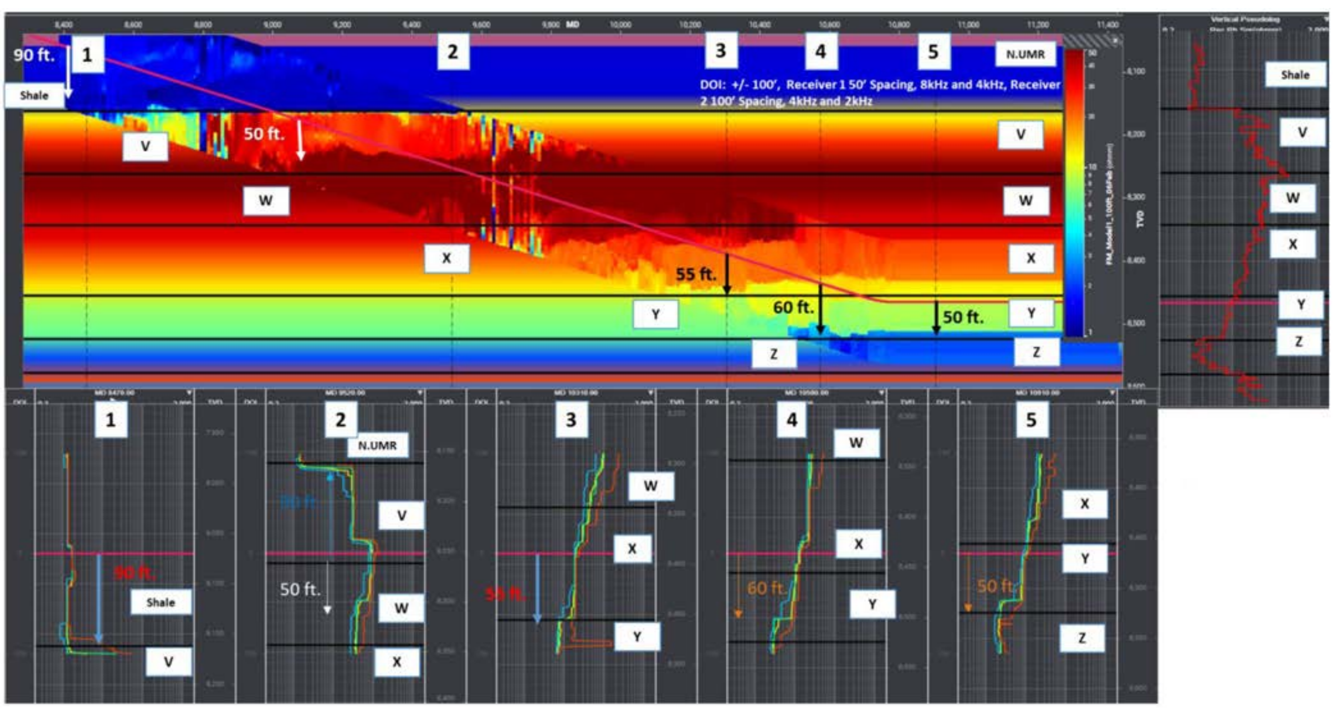

Figure 16—Inversion canvas across the 8-1/2" landing and 6" horizontal sections using offset well 7-01V. Vertical logs represent slices through the canvas

In the 8 ½" modeling section and while the wellbore is located in a shale formation, the top of limestone layer V is identified 90 ft TVD below the 80° inclination borehole. The higher resistivity in layer W is detected 50 ft TVD below the wellbore when it enters resistive layer V. While the tool is in the middle of layer X, the top of layer Y is identified at a 55 ft TVD distance. While traveling near the base of the layer X, the water flooded zone in layer Z is detected at approximately 60 ft TVD before landing in layer Y. After landing the well path in layer Y, the 6" inversion modeling detects the decrease in resistivity in layer Z approximately 50 ft TVD below wellbore but is unable to detect shallower layers due to the low resistivity contrast (Figure 16).

Prejob Inversion Modelling Using Offset Well 8-01V

Offset well 8-01V displays a significant resistivity decrease in layer W and upper layer X due to injection water encroachment. A resistivity increase at bottom of layer X is unswept oil followed by flooded interval at the top of layer Y. The resistivity profile of this offset well is quite different than earlier drilled offset wells selected for prejob inversion modeling.

Ultra-Deep resistivity modeling was conducted using 2 receiver subs for the both 8 ½" (landing section) and 6" (horizontal section) (Figure 18). The spacing between transmitter and receiver sub 1 is 50 feet while the spacing between transmitter and receiver sub 2 is 100 feet. The operating frequency combination used for the modeling was 4-8 kHz in short spacing and 2-4 kHz for long spacing. The depth of inversion was 100 feet (up and down).

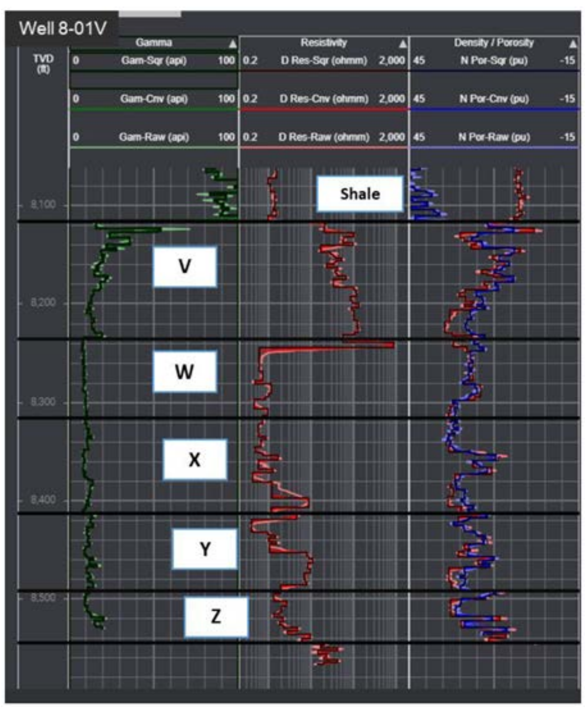

Figure 17—Basic Logs of offset well 6-01V (T1: Gamma Ray, T2: Resistivity, T3: Density-Neutron).

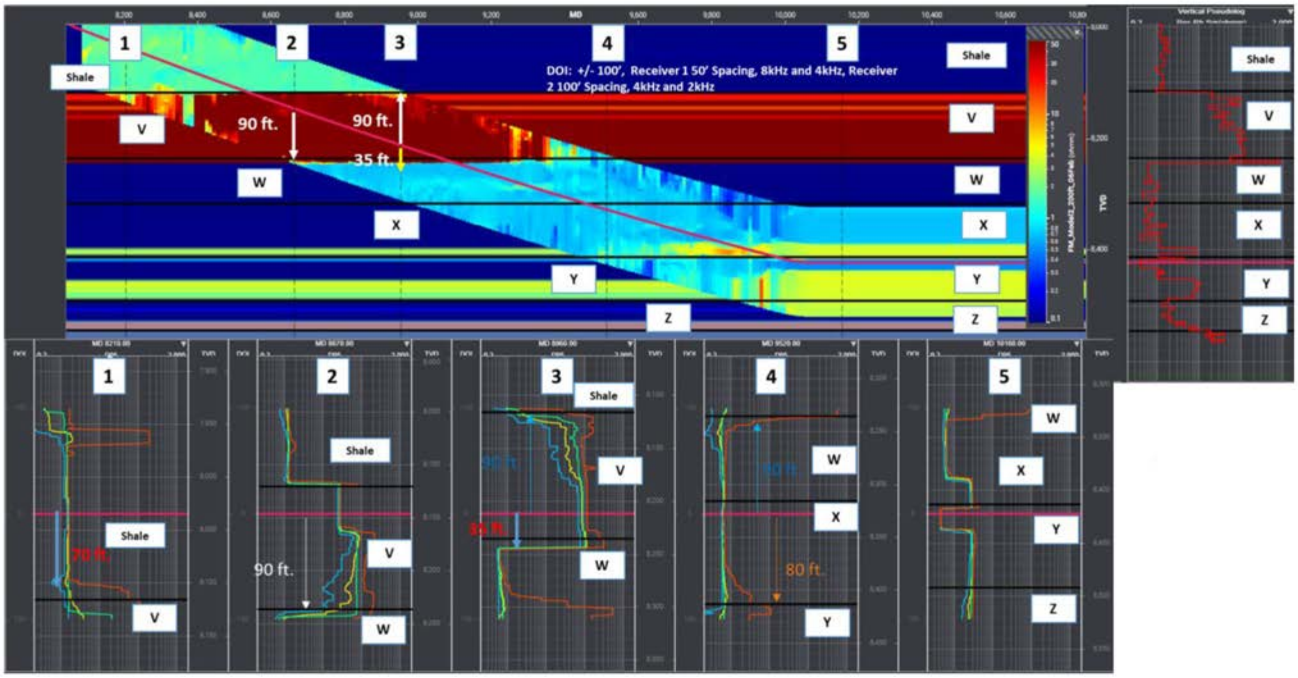

Figure 18—Inversion canvas across the landing and horizontal sections and Vertical logs to represent slices through the section. Offset well 8-01V

In the 8 ½" modeling section with the wellbore located in shale formation, the top of layer V is detected 70 ft TVD below the wellbore. With the wellbore located in resistive layer V, the water flooded zone in layer W is detected approximately 90 ft TVD below wellbore with a hole inclination of 80°. As per 6" modeling, after landing the well path in layer Y, the inversion couldn't resolve the top zones due to the poor resistivity contrast.

As per prejob inversion results using three offsets wells, it is clear for this trial well location, injection water encroachment is detectable at a desired distance from the wellbore while landing and can also be mapped while drilling 6" horizontal hole. Ultra-Deep Directional EM LWD tool should be avoided in 6" section when there is a gradational resistivity profile.

Observations

- Prejob Ultra-Deep resistivity inversion models and canvasses conducted for the trial well targets identify resistivity changes a significant distance from the well bore. This is useful in both well placement and injection water movement geomapping. Even where the resistivity profile is gradational, modelling shows that significant changes in the resistivity can be tracked allowing the wells position within the stratigraphy to be identified.
- Inversion results along with resistivity uncertainty profiles deliver distance of reliable detection of water front or formation boundaries.
- Plotting formation anisotropy and dip uncertainties would further help to access the inversion results.
- The objective of far distance water movement identification in most scenarios is achievable with the vendor's Extra Deep Directional LWD Resistivity tool used in combination with standard triple combo LWD tools.

Way Forward

- Further prejob inversion modelling to ensure ranking and choosing appropriate future trial wells using all available geological and petrophysical data, designing acquisition string, operating frequency and select real-time processing parameters etc.
- Significant advancements have been made in 1D, 2D & 3D inversion techniques. With ever increasing computation resources real-time 2D or 3D inversions may be possible. This would allow both vertical and lateral changes to be displayed for geosteering.
- All data will be incorporated into geosteering modelling software to combine the information from seismic, LWD image logs, LWD logs and the Ultra-Deep resistivity data.
- The detected reliable water front boundaries will be incorporated in static and dynamic model in order to improve understanding on water front encroachment areas, identification of bypass oil areas, improving well spacing and trajectories, optimize and operate proactive smart well completion design to enhance cumulative oil production and improve recovery factors.

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Nomenclature

DOD = Depth of reliable detection, (Maximum distance from the wellbore at which a log including its inversion processing can accurately resolve boundaries for a given set of measurements)
LWD = Logging While Drilling
MD = Measured Depth
TVD = True Vertical Depth
FT = Foot

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